# A Scalable Framework for Writer-Independent Handwriting Verification and Writer Authentication

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***Abstract*—Handwriting is a unique behavioral biometric which captures the neuromuscular and cognitive attributes of a person. We propose a text-independent handwriting verification system, consisting of deep feature extraction and person-specific classification by a machine learning system, called Biometrics-ML. To process handwritten document images, the grayscale conversion, noise reduction, adaptive thresholding and segmentation process are applied first to obtain localized handwriting patches. We gather deep embeddings for each segment using AlexNet and train one-versus-all binary classifiers using feature scaling, Principal Component Analysis (PCA) and hyperparameter optimization. At runtime, the predictions at each segment are combined using the majority approach and the overall prediction is verified and given a confidence score. The experimental results show the ability of the proposed framework to accurately verify real-time genuine hand-writing samples in both desktop and web-based applications and to successfully reject real-time impostor handwriting samples. The proposed system yields a viable, scalable and explainable solution to handwritten biometric authentication.**

***Index Terms*—Behavioral biometrics, handwriting verification, writer authentication, deep learning, AlexNet, machine learning.**

## I. Introduction

Handwriting has been known to be a behavioral trait that is individualistic and a reflection of the neuromuscular coordination, cognitive processes and personal writing habits of an individual. The formation of a stroke, slant, spacing, character structure, and the rhythm of writing makes handwriting very individual. Such difference has led to the use of handwriting widely in forensic identification, writer identification, signature verification and psychological assessment. Handwriting is a biometric modality that has become a popular option as a non-invasive alternative to physiological biometric modalities like fingerprints, iris scans and facial recognition, because of the growing demand for secure, user-friendly authentication methods.

Handwriting authentication has found novel opportunities and problems, due to the increasing reliance on digital documents and remote document verification systems. There are many factors that can cause handwritten samples to be captured under different conditions such as lighting, scanning quality, paper texture, and writing instruments. Moreover, an individual may have natural variations in his or her handwriting from one writing session to another. These factors create a great deal of complexity in the verification process. The subtle structural and stylistic features that are necessary to achieve high accuracy rate of genuine vs. impostor handwriting can not be extracted by traditional handcrafted techniques.

The recent development of machine learning and deep learning has greatly assisted the modeling of handwritten word patterns. Geometric, dynamic, pressure and muscle activation features derived from handwriting were found to have high classification accuracy in detecting and grading PD by Cascarano et al., which shows that handwriting features have high discriminative power for biomedical and biometric applications [9]. Likewise, Hasan et al. suggested a person identification system based on handwriting data collected from digital pen-tablet and a hybrid of machine learning and deep learning models and obtained an accuracy rate of 99.07% which proved that deep feature representations were very effective for writer-specific identification [3]. The above studies have shown that handwriting is a wealth of behavioral data that can be successfully captured and analysed using modern AI techniques.

In spite of these advances, there are several issues that have not been addressed. Most of the current handwriting verification systems are based on pre-designed text or signatures, which can not be applied in the real-world situation. Some methods make use of special equipment like digital tablet for obtaining the dynamic writing information, which increases the system cost and complexity. Others use end-to-end deep learning models, however, which require massive amounts of data and high computing power. Moreover, many of them do not return results that are easily understood and provide no confidence measures, which makes them of limited use for security-sensitive applications.

To overcome these limitations, this paper introduces a person-specific machine learning-based text independent handwriting verification system called Biometrics-ML using deep convolutional feature extraction and person-specific machine learning classifiers. The proposed system includes image preprocessing, adaptive segmentation, and embedding generation through AlexNet that can be used to process handwritten documents that are scanned or photographed. Individuals are trained separately with balanced dataset, feature scaling and dimensionality reduction for a one-versus-all binary classification. At runtime, the system analyzes the prediction for each segment, and uses majority voting to decide if the handwriting is from the selected person, and a confidence score to show the confidence level of the decision. It's deployed as a desktop and web application, making it possible to use it in practical applications such as document authentication, forensic analysis, examination security, and behavioural biometric access control.

## II. Literature Survey

Much research has been done on the use of handwriting as a behavioral biometric to identify neurological disorders, predict demographics, analyze personality, and for user authentication. Gornale et al. gave a detailed survey on handwritten signature biometric analysis for the evaluation of neurological disorders such as Parkinson's disease (PD) and Alzheimer's disease (AD). They reviewed various datasets, feature extraction methods and machine learning classifiers that are commonly used, and highlighted the importance of handwriting as a non-invasive biomarker that can capture fine motor impairments and cognitive decline. Some challenges for the open survey were also found: Limited datasets, lack of standardization, and the need for robust multimodal approaches [1]

Dargan et al. suggested a handwriting based gender classification system with 200 writers of offline Gurumukhi script samples. They used classifiers such as K-NN, Decision Trees, Random Forest, and AdaBoost in conjunction with zoning, diagonal, peak extent and transition features. Experimental results indicated that the hybrid feature extraction technique was very effective in the classification process, achieving the highest classification accuracy of 94.6% for handwriting as a behavioral feature for demographic inference, proving the handwriting as a behavioral feature is viable for demographic inference [2]. Hasan et al. proposed a handwriting analysis system for person identification using a digital pen-tablet based system. A total of 91 statistical, kinematic, spatial and composite features were extracted from the handwriting samples and fourteen machine learning and seven deep learning models were applied. Optimization of performance was done using the feature selection techniques like ANOVA-F and PCA. The best accuracy of 99.07% was obtained by CatBoost model, as dynamic handwriting signals proved to be effective for reliable and cheap biometric authentication among all models [3].

One personality prediction study using handwriting used image preprocessing and Convolutional Neural Networks (CNNs) to identify handwriting characteristics including slant, spacing and writing pressure. The system was able to identify personality traits such as optimism, shyness, and emotional balance by analyzing handwritten images through various stages, such as thresholding, feature extraction, and trait mapping. This work showed the possible application of automated graphology in the fields of recruitment, matrimonial matching, etc. [4]. Joshi et al. suggested a machine learning method for handwriting personality traits detection using graphological features like margins, baseline direction, slant and characteristics of the T-bars. To enhance adaptability and prediction accuracy, the authors used a K-Nearest Neighbor (KNN) classifier with incremental learning. Their system was designed to be used by graphologists, in order to reduce the time needed to analyse the graph, while maintaining the interpretability, thus allowing large-scale behavioral assessments [5]. Gornale et al. proposed a gender classification system based on signature features extracted by feature fusion of Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG) and statistical descriptors. The study employed a handwritten signature data set of 4790 signatures and employed three classifiers namely KNN, Decision Tree and Support Vector Machine for classification of gender. The SVM model has a perfect accuracy of 100% which proves the discriminative power of texture and structural features in behavioural biometrics and forensic document analysis [6].

A comparative study of machine learning and deep learning techniques for writer identification in English and Arabic handwritten database was performed by Durou et al. The AlexNet convolutional neural network was compared with traditional classifiers like Support Vector Machine and K-Nearest Neighbors. The results showed that deep learning models automatically learned highly discriminative features, and that they outperformed the conventional methods, particularly in the multilingual and complex handwriting cases [7]. In this novel approach, the authors of this paper, Aleman et al., introduced a new methodology, which is a combination of robotic modeling and machine learning to classify gender from handwriting. The study simulated handwriting movements with a robotic arm and estimated kinematic and dynamic signals and extracted features by applying the Linear Predictive Coding and Singular Spectrum Analysis. Experiments conducted on healthy subjects and patients with Parkinson's disease showed that there are significant motor differences between healthy and PD patients and that robotic handwriting modelling is a promising approach to cognitive biometrics and neurodiagnostics [8].

Cascarano et al. studied handwriting abnormalities for the purpose of assessing and grading Parkinson's disease. They were able to collect geometric, dynamic, pressure, and surface electromyography (sEMG) signals from their system as the participants wrote on a graphic tablet. Artificial Neural Networks were then applied for the classification of healthy subjects from PD patients and the classification of the severity of the disease. The proposed method has achieved more than 90% accuracy, which shows that the multimodal handwriting features can be well-used for clinical decision making [9]. Zaibi and Bezine suggested a learning disabilities detection system, based on handwriting analysis and machine learning, for dyslexia, dysgraphia and dyscalculia learning disabilities. The study was based on the Handyg23 dataset and included spatial, temporal and kinematic descriptors extracted from the dataset using the theory of Beta-Elliptic segmentation. A number of classifiers were tested and the Gradient Boosting classifier was found to be the most accurate with a score of 99%. The results show that handwriting analysis can be a powerful and non-invasive educational screening and timely intervention tool [10].

## III. Methodology

### A. System Overview and Architecture

The proposed handwriting verification system architecture is a multi-stage pipeline, which involves image preprocessing, deep feature extraction, machine learning based classification and person verification at runtime. Initially, person-wise handwriting image datasets are collected and organized according to individual identities. The raw handwritten document images are then fed into an image preprocessing and segmentation module that processes the images to create meaningful handwriting patches through grayscale conversion, noise reduction, contrast normalization, binarization and sliding-window-based segmentation. These pre-processing steps improve the quality of the image and segment out the handwritten regions, which improves the performance of the feature extraction and classification tasks in the downstream process.

After preprocessing, the segmented handwriting patches are fed into the deep feature extraction module that is based on AlexNet. At this step, each handwriting segment is converted to a high dimensional embedding vector that captures the semantic and structural properties of the handwriting. The extracted embeddings are outputs of different layers of AlexNet, allowing the system to learn low level and high level handwriting features. These feature vectors are then saved in a structured vector repository in the parquet format together with the person labels, to facilitate retrieval and training of the model.

The feature repository thus generated is then used in the construction of binary datasets and training of classifiers. The system builds a balanced binary classification dataset for each individual, with positive samples being the individual's own and negative samples being from all other individuals. To reduce the dimensionality and to enhance the generalization of the classifiers, feature scaling and Principal Component Analysis (PCA) are used. Cross validation and hyper parameter optimization are performed to train and test multiple machine learning models such as Random Forest, Support Vector Machine, XGBoost, Logistic Regression and Naive Bayes. The best model is chosen for each individual according to evaluation metrics like accuracy, precision, recall and F1-score and then serialized as a pretrained .pkl model for deployment.

In the run time inference, the user enters a handwritten image and the desired person ID into the application interface. The image to be processed is preprocessed and segmented using the same pipeline as the one used during training, which guarantees that the training and testing environments are consistent. Each segment generated is then encoded as AlexNet embeddings and fed to the respective pretrained AlexNet classifier model. The system makes predictions for each segment, and uses a majority voting approach to decide the final verification result. The framework predicts and outputs the confidence score and whether the handwriting is from the specified person or not, based on the aggregated predictions and the confidence score. This end-to-end architecture allows for the accurate, scalable and automatic handwriting verification using a hybrid deep learning and machine learning approach.

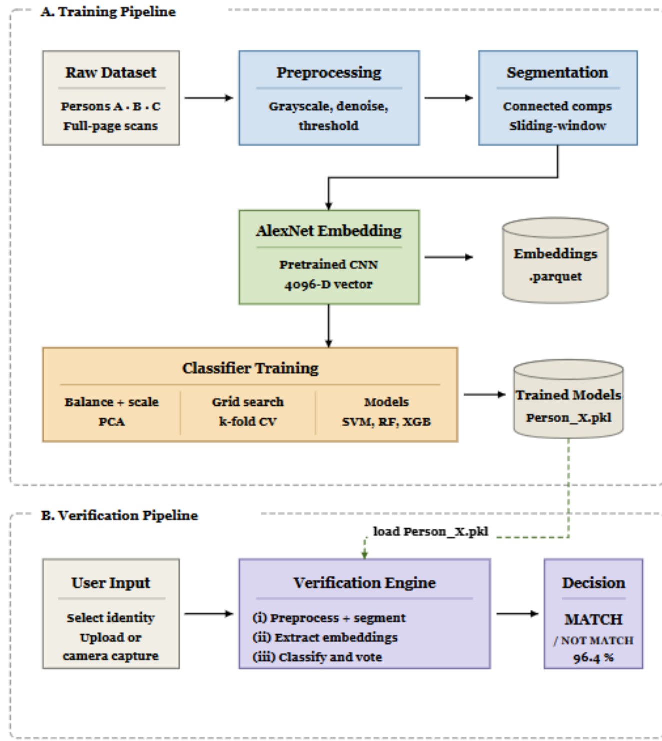


Fig. 1: End-to-end architecture of the Biometrics-ML framework for text-independent handwriting-based biometric authentication.

### B. Dataset Creation and Image Preprocessing

The creation of the dataset and pre-processing of images are an essential part of the proposed handwriting verification system. The main goal of this phase is to convert the handwritten document images into structured and standardized representation for deep feature extraction and classification using machine learning techniques. The overall pre-processing pipeline helps to reduce noise, illumination, background differences and handwriting irregularities as much as possible prior to feature embedding generation.

The dataset is structured in a person-wise manner with each person having several handwriting samples in different folders. The folders represent different persons and each folder has handwritten document images scanned under different writing conditions. The structured organization allows the framework to produce person-specific binary classification datasets at the training time. The Preprocessing Pipeline performs the preprocessing of all person folders and each handwriting image is preprocessed separately, then segmented handwriting patches are created for feature extraction.

The first step in the raw handwriting images is to convert them to grayscale. During this process the multichannel color image is transformed to a single channel image containing only brightness information, known as an intensity image. The handwritten stroke structure is retained and the computational complexity is reduced by converting the color image to a grayscale image because color information is not necessary for handwriting analysis. This simplification allows downstream processing to be done efficiently and the segments are more consistent. After grayscale conversion, the system applies a Non-Local Means (NLM) filtering to reduce the noise. The handwritten document images may suffer from scanning artifacts, random disturbances of the pixels, ink spread and paper texture noise that may have a negative impact on the segmentation and feature extraction process. The NLM filtering algorithm is able to suppress unwanted noise while retaining edges and texture details of handwriting. The NLM method preserves the continuity and sharpness of handwriting strokes, which is not the case for traditional averaging filters, and thus enhances the image quality of the processed image.

Denoising, contrast normalization and adaptive binarization are applied to enhance the visibility of handwritten characters and foreground-background separation. The proposed framework uses adaptive Gaussian thresholding, which means that the value of the threshold is computed locally for the image regions, rather than globally. This way, the preprocessing pipeline remains robust in the face of non-uniform illumination as well as different backgrounds on documents. The binary inverse thresholding operation is performed to make the handwritten regions stand out as foreground pixels and the background as background pixels, and thus to obtain a clean binary mask representation of the handwritten content.

The obtained binary image is then fed into the handwriting segmentation module. Here Connected Component Analysis (CCA) is used to segment the handwriting areas and remove the insignificant noisy components within area limits. A statistical analysis is then carried out to determine the characteristic dimensions of the handwriting, e.g. median stroke height or median component height. Based on these estimated dimensions, a sliding-window segmentation strategy is then used over the document image, to produce localized handwriting patches. A patch of handwriting image is extracted from each sliding window and patches with no information or with little information are discarded. By segmenting the handwritten page into distinct segments, the framework can focus on specific handwriting patterns within each segment, which helps to represent features and improve the robustness of classification.

The handwriting patches are segmented, resized and standardized after segmentation and then passed to the deep feature extraction stage. Each of these processed segments is then mapped to a high dimensional feature vector that captures the semantic and structural properties of the handwriting by the AlexNet embedding model, which is the main input to the model. The resulting feature vectors are then saved in a parquet repository, which serves as the input data set for the next steps in the machine learning training and verification process, namely the person labels.

### *C. Handwriting Segmentation and Feature Extraction*

The analytical phase of the proposed handwriting verification system is the handwriting segmentation and feature extraction. The main goal of this stage is to separate meaningful handwriting regions from the preprocessed document image and represent them in a compact form that can be used as numerical representation for machine learning based classification. The framework conducts localized handwriting analysis instead of analyzing the entire handwritten page as a whole, in order to better capture details of writer-specific characteristics.

The first step in segmentation is Connected Component Analysis (CCA) in which the binary handwriting image is scanned to find the handwriting regions and remove small noisy artifacts by applying area-based filtering constraints. Then, statistical analysis is carried out to estimate characteristic handwriting dimensions and then dynamically calculate the window size of segmentation. A segmentation scheme based on sliding windows is then applied to the document image to produce a set of handwriting patches in the image. Valid handwritten regions are kept for further processing and invalid or empty regions are discarded.

Each handwriting patch extracted after segmentation is then put through the deep feature extraction module based on AlexNet. The aim of this stage is to convert the handwriting segments into embedding vectors of high dimension, which represents structural and semantic handwriting characteristics. As opposed to handcrafted features, AlexNet embeddings learn complex stroke patterns, writing styles, and information at the level of texture directly from the handwriting images, which leads to better quality of feature representation.

The framework learns embeddings from several layers of the AlexNet architecture, such as the fully connected layers and the softmax output layers, to get rich handwriting representations. These are the handwriting pattern's compact mathematical "fingerprints" and are stored in a repository in parquet format together with person labels. The embeddings generated in this step are used as the main input to the next model training and handwriting verification steps of the proposed system.

### *D. Feature Vector Construction and Model Training*

The embeddings extracted from the handwriting are transformed to structured datasets for supervised machine learning-based verification in the feature vector construction and model training stage. All handwritten embeddings generated by the AlexNet are saved in a person label and segment information along with the embeddings in a parquet-based repository. The centralized feature storage system is the key component of this repository for efficient retrieval and training of classification models.

The framework builds a binary classification dataset for each individual in the data set, in a one-versus-all approach. The handwriting feature vectors of the target person are considered as positive samples and the feature vectors of other persons are considered as negative samples. The system employs a balanced dataset generation technique to tackle the class imbalance problem, where the number of negative samples is kept close to the number of positive samples. This is a balanced representation to avoid model bias towards majority classes and to enhance verification performance.

The feature vectors are subjected to some pre-processing before the model is trained. The data distributions are normalized using StandardScaler and the dimensionality of the features is reduced using Principal Component Analysis (PCA) while retaining most of the variance in the data. Dimensionality reduction reduces the number of computations, decreases overfitting and enhances the generalization ability of the classification models.

The framework trains several machine learning classifiers such as Random Forest, Support Vector Machine (SVM), Logistic Regression, Naive Bayes and XGBoost models. Hyperparameter tuning is done with GridSearchCV and Stratified K-Fold cross validation to get optimum performance. This training method allows for a systematic comparison of various classifiers and provides strong and repeatable model evaluation for various training folds.

All the classifiers are tested after training with various measures like accuracy, precision, recall and F1 score. The best classifier (with the highest test accuracy) for a certain individual is picked and then serialized into a pre-trained .pkl model file for deployment. These are then used at run-time for final handwriting verification and prediction.

### *E. Performance Evaluation Metrics*

The performance evaluation stage is in charge of evaluating the effectiveness and reliability of the proposed handwriting verification framework. The system is trained with several machine learning classifiers, and then tests each one to determine how well it can correctly classify handwriting samples from the target person and those from other people. The evaluation process is performed on a separate test set and training set to provide unbiased evaluation of the model and proper generalization.

Several metrics like accuracy, precision, recall and F1 score are used to comprehensively analyze the performance of the classifiers. The accuracy indicates the percentage of all handwriting samples correctly classified, and precision indicates the percentage of all handwriting samples correctly predicted when they are all positive. Recall is the ability of the model to correctly identify authentic handwriting samples of the target person. The F1-score is a balanced measure that takes into account both precision and recall, and is especially suitable for binary verification systems that may suffer from class imbalance.

The framework is used to compare several classifiers like Random Forest, SVM, Logistic Regression, Naive Bayes and XGBoost and find the best performing classifier for handwriting verification. The model that performs best in the test and generalizes best is chosen as the final verification model to be deployed.

### *F. Runtime Inference*

The real-time handwriting verification is done at the runtime inference and verification stage, which involves the use of the pretrained classification models. Inference is performed when the user enters a handwritten document image and the identity of the target person in the application interface. The system then loads the corresponding pretrained model corresponding to the selected person and starts the verification pipeline.

The input handwriting image is preprocessed and segmented in the same way as during training to make sure that the same processing and segmentation are performed in both training and test phases. The image of the document is converted to grayscale, denoised, normalized, and segmented into patches of handwriting using the sliding-window segmentation strategy. The generated segments are then fed into the AlexNet feature extraction module to obtain deep embedding vectors of the handwriting features of the input sample.

The extracted feature vectors are then fed into the pretrained classifier model for segment-wise classification. Each handwriting segment is analyzed separately, and the framework uses a majority voting approach to combine the individual handwriting verification results and obtain the final handwriting verification result. An additional confidence score is also calculated using the percentage of majority predictions to the total number of predictions. Lastly, the system can provide the result of whether the handwriting is from the specified person or not, which can realize the accurate and automatic handwriting verification.

## IV. Results And Discussion

### *A. Pre-processing and Segmentation Results*

The pre-processing pipeline proposed in this paper was able to improve the quality of handwritten document images before feature extraction and classification. Grayscale conversion, noise reduction, adaptive thresholding and contrast normalization were among the operations that greatly enhanced the clarity of handwritings and reduced background noise and varying illumination levels. The preprocessing stage was able to create clean binary handwriting representations that enhanced the effectiveness of downstream segmentation operations.

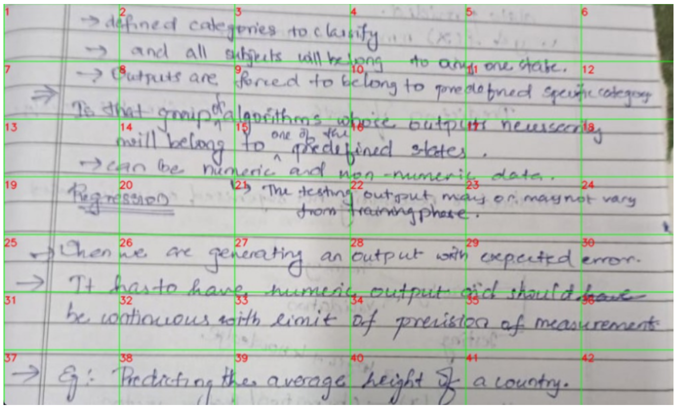

Fig. 2: Segmentation

The segmentation module achieved good results in the segmentation of localized handwriting regions by combining Connected Component Analysis (CCA) with a sliding-window segmentation strategy. The framework did not process the entire handwritten page, but rather obtained several informative handwriting patches from various regions of the page. This localized segmentation method helped the system to better capture the writer-specific stroke patterns, spacing characteristics and handwriting structures.

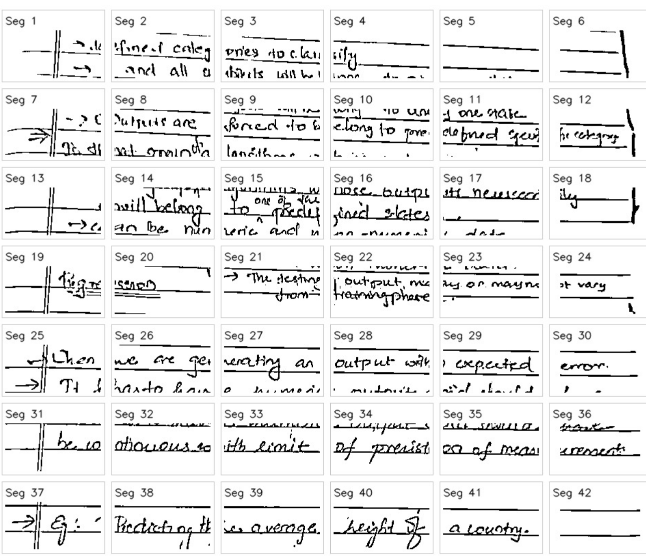


Fig. 3: Grayscale image

The handwriting segments generated were then fed into the AlexNet feature extraction pipeline to extract deep embedding vectors. The feature diversity was greatly enhanced and more training samples were obtained from each document image with the segment-level analysis. The segmentation-based pipeline thus played a significant role in boosting the robustness and generalization of the handwriting verification framework.

### B. Model Performance Comparison

The suggested framework trains and tests various machine learning classifiers for person-wise handwriting verification task. Balanced binary handwriting samples were formed with positive and negative samples for each person and the deep feature embedding vectors obtained from AlexNet were used to train the classifiers. The framework is based on one versus all verification approach, which entails training a binary classifier for each individual in the data set. Different classifiers such as Random forest, SVM, Logistic regression, Decision Tree, XGBoost, Gaussian Naive bayes, KNN and MLP are applied to comparative evaluation to find out the best performing model in handwriting verification.

The classification results showed that the proposed framework had the ability to classify handwriting samples effectively whether they are similar or dissimilar. Logistic Regression and XGBoost outperformed the other models in terms of testing accuracy, with logistic regression showing the best performance with 91.60%, and XGBoost also demonstrating the best performance in terms of the recall score with 95.38%, in which it performed well at correctly identifying genuine handwriting samples (see Table X). Likewise, the test accuracy of the MLP classifier was 90.8% and precision-recall values were balanced overall. Gaussian Naive Bayes, on the other hand, had relatively low testing accuracy because of its oversimplified assumptions about the probability function. The results obtained show that the capability of the classifiers to discriminate has improved significantly by using deep embedding based hand-writing representations. The framework

TABLE I: Performance Comparison of Different Machine Learning Classifiers on One Sample

| Model | Train Acc | Test Acc | F1 Score | Precision | Recall | FAR | FRR | EER |
|---|---|---|---|---|---|---|---|---|
| Random Forest | 0.9885 | 0.8855 | 0.8855 | 0.8906 | 0.8769 | 0.1061 | 0.1231 | 0.0019 |
| SVM | 0.9712 | 0.8473 | 0.8472 | 0.8689 | 0.8154 | 0.1212 | 0.1846 | 0.0028 |
| Logistic Regression | 0.9885 | 0.9160 | 0.9160 | 0.9091 | 0.9231 | 0.0909 | 0.0769 | 0.0012 |
| Decision Tree | 0.9885 | 0.8550 | 0.8545 | 0.8966 | 0.8000 | 0.0909 | 0.2000 | 0.0939 |
| XGBoost | 0.9885 | 0.9160 | 0.9159 | 0.8857 | 0.9538 | 0.1212 | 0.0462 | 0.0007 |
| Gaussian Naive Bayes | 0.9002 | 0.8321 | 0.8316 | 0.8644 | 0.7846 | 0.1212 | 0.2154 | 0.0790 |
| KNN | 0.9175 | 0.8626 | 0.8626 | 0.8615 | 0.8615 | 0.1364 | 0.1385 | 0.0021 |
| MLP | 0.9885 | 0.9084 | 0.9084 | 0.9077 | 0.9077 | 0.0909 | 0.0923 | 0.0014 |

also conducted statistical analysis on the resulting metrics (mean, standard deviation, minimum and maximum values) of various classifiers. The proposed framework obtained an average testing accuracy of 85.43% and an average testing F1 score of 86.06% as shown in Table Y. The evaluation also showed low Equal Error Rate (EER) values, which means it has a satisfactory level of verification accuracy, meaning the chance of a false acceptance rate and false rejection rate is minimized. Moreover, the low standard deviation values acquired on various measures validate the uniformity and stability of the proposed framework for handwriting verification based on segmentation. The experimental results show that the segmentation-based handwriting analysis, AlexNet feature embeddings, and person-specific machine learning classifiers is effective hybrid handwriting verification approach. Localization of handwriting segments led to the creation of a more diverse set of features and to the creation of a more robust set of features for classification purposes, and feature

TABLE II: Statistical Analysis of Evaluation Metrics

| Metric | Mean | Std | Min | Max |
|---|---|---|---|---|
| Train Accuracy | 0.960371 | 0.057121 | 0.673820 | 1.000000 |
| Test Accuracy | 0.854312 | 0.096200 | 0.600000 | 1.000000 |
| Train Recall | 0.974217 | 0.047615 | 0.629630 | 1.000000 |
| Test Recall | 0.898474 | 0.096162 | 0.571429 | 1.000000 |
| Train Precision | 0.951097 | 0.067731 | 0.630573 | 1.000000 |
| Test Precision | 0.834897 | 0.118012 | 0.571429 | 1.000000 |
| Train F1 Score | 0.961820 | 0.054657 | 0.722628 | 1.000000 |
| Test F1 Score | 0.860657 | 0.089803 | 0.571429 | 1.000000 |
| Train FAR | 0.053560 | 0.078363 | 0.000000 | 0.500000 |
| Test FAR | 0.188249 | 0.149907 | 0.000000 | 0.625000 |
| Train FRR | 0.025783 | 0.047615 | 0.000000 | 0.370370 |
| Test FRR | 0.101526 | 0.096162 | 0.000000 | 0.428571 |
| Train EER | 0.039671 | 0.057197 | 0.000000 | 0.326923 |
| Test EER | 0.144888 | 0.095731 | 0.000000 | 0.401786 |

scaling, PCA dimensionality reduction, and hyperparameter optimization were used to further increase the generalizability of the model. The trained models were then picked up as .pkl files and used in the runtime inference for handwriting verification in real-time.

### C. Runtime Verification

The proposed Biometrics-ML framework was validated by real-time runtime verification, both in the form of a desktop based graphical user interface and a web application. The aim of this experiment was to show the working of the system when it is given authentic and fake handwritten samples. In both, the user chooses one of the trained models, uploads a handwriting image and starts the verification process. The system then preprocesses the image, segments the handwriting into local patches, extracts deep feature embeddings, and classifies the handwriting on a segment by segment basis. The final decision is made by majority voting and a confidence score is generated depending on the ratio of the votes that voted for the majority class.

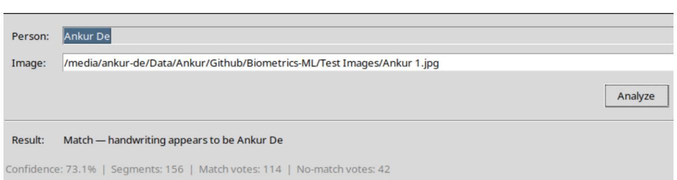


Fig. 4: Matching Handwriting

The handwriting system was able to correctly identify the handwriting sample as written by the selected person when a sample handwriting was compared against a model handwriting trained for the selected person. The result of the verification was returned as Match with a confidence score of 73.1%. The input image was divided into 156 handwriting patches, among which, 114 were classified as matching and 42 were classified as non-matching. The system gave a positive authentication result because most of the segments were in favor of the selected writer.

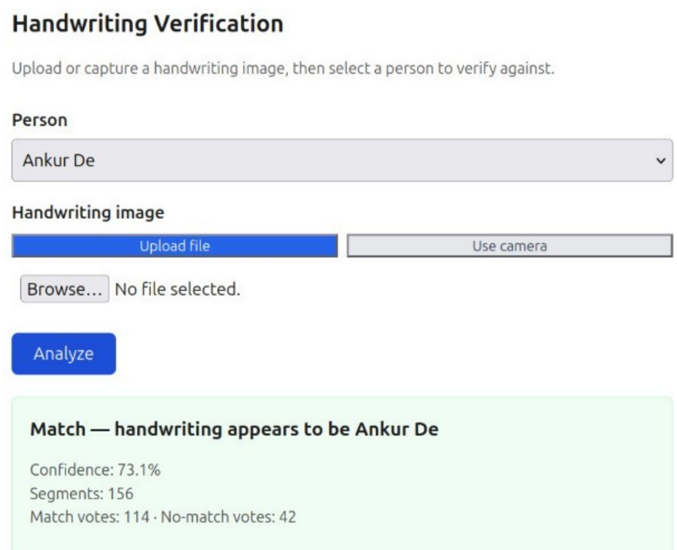


Fig. 5: Web based handwriting Verification interface for matching sample

This handwriting sample was then compared to the model trained for Trisha. In this instance the system was able to correctly determine that the handwriting was not the selected person's and issued a No Match result. The confidence score was 98.1%, and 3 segments were wrongly identified as matching segments, while 153 segments were identified as non-matching segments. The result shows that the model has a good rejection capability of handwriting samples from other persons.

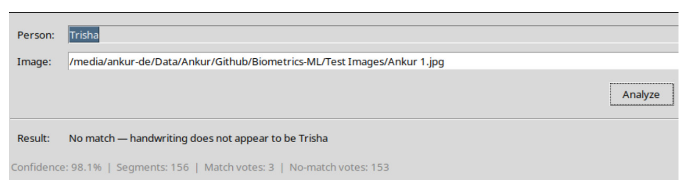


Fig. 6: Non-matching Handwriting

When the handwriting image was compared to the original writer, the same result was obtained with the web-based application. The system returned a Match decision with the same stats, which is exactly what the web interface does as well. This consistency helps to ensure that the underlying inference engine and its implementation in various user interfaces are correct. In the same manner, the system was able to correctly return a No Match response with a 98.1% confidence score when the same image was matched against the model of another writer using the web application. The high rejection confidence suggests that most handwriting segments were rejected as being significantly different from the handwriting patterns learned by the selected writer.

The runtime verification results show that the proposed Biometrics-ML framework is reliable and interpretable handwriting-based authentication in real-time. The system was

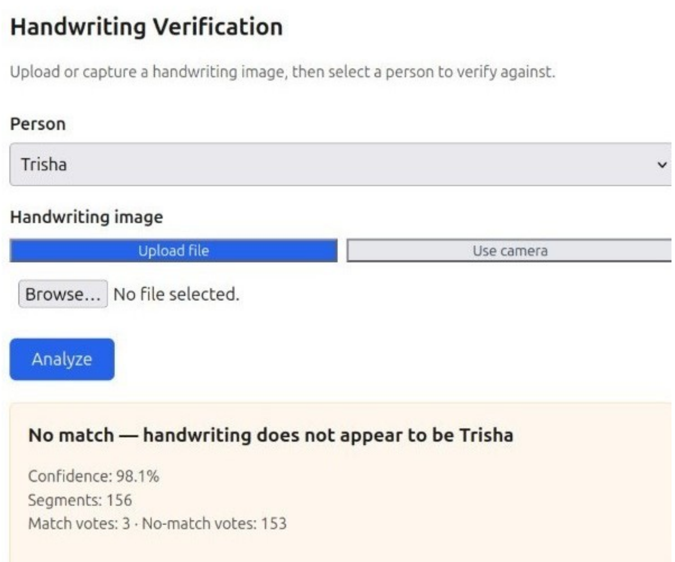


Fig. 7: Web based handwriting Verification interface for non-matching sample

able to accept genuine handwriting samples and reject impostor handwriting samples with success. The uniform results on the desktop and web platforms also highlight the strength, versatility and usability of the framework for behavioral biometric authentication.

## V. Conclusion

Handwriting based biometric authentication has shown itself to be a promising and non-invasive method for the identification of an individual by examining the writing style that they use. The proposed Biometrics-ML framework shows that combination of deep convolutional neural network (DCNN) embeddings and person-specific machine learning (ML) classifiers can provide reliable, scalable and interpretable text-independent handwriting verification.

The proposed pipeline features a multi-stage approach: image pre-processing, handwriting segmentation, feature extraction using AlexNet, feature scaling, dimensionality reduction and the training of one-versus-all binary classifiers. The system processes localized handwriting patches instead of the whole document image to learn writer-specific structural and stylistic properties more discriminatively. The extracted deep embeddings are used to represent the handwriting sample as strong features, which helps to accurately classify handwriting samples from multiple individuals.

The effectiveness of the proposed approach was validated in experimental evaluations in real-time verification scenarios. The system was able to correctly classify true handwriting samples and correctly classify impostor samples using a max voting approach at the segment level. In the experiments conducted during runtime, the correct handwriting sample was correctly verified with a confidence of 73.1% while it was rejected with a confidence of 98.1% against a different handwriting sample. The results indicate that using a hybrid approach of deep learning and machine learning is a potent method for text-independent handwriting authentication.

The trial of both Web and desktop interfaces further demonstrated the practicality of the framework. The results of the verification were consistent across the two platforms and interpretable answers such as confidence score, number of segments and vote distribution were obtained. This illustrates the transferability, applicability, and stability of the proposed framework for application in the real world.

The proposed method is practical, low cost, and convenient and has the potential to be used in document verification, forensic handwriting analysis, examination security, and access control systems. The future directions include expanding the handwriting database, using more complex architectures (e.g., ResNet or Vision Transformers), integrating dynamic handwriting parameters (pen pressure and velocity), and implementing and evaluating a dynamic open set handwriting verification system for enhanced accuracy and scalability.

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